



Tech Screening

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Chapter 1. Assignment

Introduction

At BIMcollab, we create powerful tools that elevate the quality and reliability of Building Information Management (BIM). Our software helps architects, engineers, and construction professionals collaborate smoothly, ensuring that building plans and models are precise and error-free. One essential part of our BIM solutions is clash detection—a critical process that identifies and resolves issues early, preventing costly mistakes before construction begins.

This technical assessment offers you a glimpse into our world: developing smart solutions to complex problems. We look forward to seeing how you approach this challenge and can't wait to discuss your insights and choices.

What We'll Be Looking At

In reviewing your solution, we'll pay close attention to your approach and thought process, especially regarding these aspects:

- **Architecture:** The logical structure and organization of your solution.
- **Testability:** How easily your solution can be tested and maintained.
- **Quality:** Your commitment to clarity, readability, and reliability.

We're not just looking at your final result—we're equally interested in understanding the decisions, trade-offs, and shortcuts you've chosen along the way.

Guidelines & Expectations

This assessment is intended as a conversation starter to better understand your approach and way of working. Here are some helpful guidelines:

- **Time limit:** Aim to spend no more than two hours. Prioritize quality over quantity.
- **Completeness:** A fully working solution isn't necessary, but providing something runnable is appreciated.
- **Quality first:** Show us your best work in the areas that matter most to you.
- **Submission:** Please reply directly to our recruitment email with your solution attached or linked. This ensures it's automatically stored in our system.
- **Questions welcome:** Feel free to reach out anytime if you need clarification or assistance.

And importantly—have fun with the assignment!

Chapter 2. Business Case

Introduction

High-quality Building Information Management relies heavily on effective clash detection. In collaborative construction projects, ensuring that buildings adhere to precise plans and comply with zoning regulations is crucial. For this assessment, your goal is to create a proof-of-concept application that detects and reports issues within provided building and site plans. You'll leverage validation and business rules to identify clashes and ensure building compliance, demonstrating how your solution can proactively address design conflicts and zoning constraints.

The Challenge

We have a site plan along with several proposed buildings. Your task is to validate whether each building fits appropriately within the site boundaries, adheres to regulatory requirements, and complies with zoning laws. Additionally, you'll need to ensure the provided input data is complete and valid.

The clash detection application will serve as a backend component and must provide a clear and usable API for integration with other applications.

Requirements

- The application must accept input data via an API, so that other systems can use the clash detection functionality as an external service.
- The application must validate the input data according to the validation rules listed below.
- The application must detect all possible clashes as defined by the business rules.
- The application must return a list of detected clashes. This output will be used by other systems to generate tasks for resolving the issues.

Example Data

An example json dataset is included in the appendix [Dataset](#), at the end of this document. Different configurations may be provided and discussed during the review. You can also find the dataset on [GitHub](#).

The dataset contains the following:

The Site Plan

Width	Length
1000 units	500 units

The Buildings

Building	Type	Width	Length	X	Y
Oakwood Academy	School	300	300	0	0
Pulse	Nightclub	200	200	0	500
Centennial Park	Stadium	700	300	400	300
Willow Residence	Residential Building	200	200	300	100
Maple Plaza	Office	150	150	250	150

Validation Rules

- All attributes of a building (name, type, width, length, x, y) must be provided.
- Building dimensions (width, length) must be positive numbers, greater than zero.
- Positions (x, y) must be numbers greater than or equal to zero.

Business Rules

General business rules:

- Buildings must be positioned fully within the boundaries of the site plan.
- Buildings cannot overlap with each other.
- Each building must maintain a minimum clearance distance of 10 units from other buildings.

Zoning business rules:

- Nightclubs must be at least 200 units away from any school.
- Residential buildings must be at least 150 units away from stadiums and nightclubs.

Chapter 3. Dataset

This is the JSON representation of the data shown earlier in the assignment.

```
{
  "sitePlan": {
    "width": 1000,
    "length": 500
  },
  "buildings": [
    {
      "name": "Oakwood Academy",
      "type": "School",
      "width": 300,
      "length": 300,
      "x": 0,
      "y": 0
    },
    {
      "name": "Pulse",
      "type": "Nightclub",
      "width": 200,
      "length": 200,
      "x": 0,
      "y": 500
    },
    {
      "name": "Centennial Park",
      "type": "Stadium",
      "width": 700,
      "length": 300,
      "x": 400,
      "y": 300
    },
    {
      "name": "Willow Residence",
      "type": "ResidentialBuilding",
      "width": 200,
      "length": 200,
      "x": 300,
      "y": 100
    },
    {
      "name": "Maple Plaza",
      "type": "Office",
      "width": 150,
      "length": 150,
      "x": 250,
      "y": 150
    }
  ]
}
```